

Alexander Bloom

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EXPERIENCE

VT BOLT

Blacksburg, Virginia

Junior Electrical Team Lead

Sep 2024 - Present

- Leadership for hardware development and electrical systems.
- Assisted in leading a team of 60 with project direction and consultation.
- Managed electrical and mechanical system documentation.
- Operated and led testing and planning for battery and controls subsystems.

Hume Research Center

Blacksburg, Virginia

Night's Watch Simulation Design Team

Sep 2024 - May 2025

- Developed MATLAB simulations for UAV/UAS flight path modeling and obstacle avoidance.
- Implemented Python-Based drone environment with VisPy for real-time visualization.
- Collaborated on AI-Integrated counter-UAV research simulations.

PROJECT EXPERIENCE

VT BOLT

Blacksburg, Virginia

A2C (Anything To CAN)

- Designed PCB utilizing an STM32 for adaptively interpreting sensors and transmitting them to a central CAN bus.
- Capable of receiving SPI, I2C, UART, and Analog Signals universally from external sensors.

Battery Testbench

- Led a team of 4 in project design and design integration.
- Designed PCB utilizing an STM32 for measuring battery cell discharge, temperature, and current draw.
- Integrated custom power management circuitry utilizing the STM32 L9963E BMIC.

Internal Wiki

- Built custom React.js frontend for the user to catalogue team knowledge for internal use.
- Built MongoDB database infrastructure backend.

EDUCATION

Virginia Tech

Blacksburg, Virginia

B.S. Electrical and Computer Engineering: Chip-Scale Integration

Graduation Date: May 2027

GPA: 3.13 / 4.0

TECHNICAL SKILLS

Technical: Schematic Design, PCB Layout, SMD Soldering, CAD, FDM 3D Printing

Software: Altium Designer, MATLAB, Fusion360, AutoCAD, MS Suite, Git/Github/Gitlab, Quartus, Microsoft Office

Programming: C, C++, Java, JavaScript, HTML/CSS, Python

ACHIEVEMENTS

2024 Technology Student Association Nationals - Structural Engineering and Design 3rd Place

2023 Technology Student Association Nationals - Structural Engineering and Design 1st Place

2022 Technology Student Association Nationals - Structural Engineering and Design 3rd Place

RELEVANT COURSEWORK

Fundamentals of Digital Systems - Embedded Systems - Computer Architecture - Data Structures & Algorithms